

Pharmaceutical Drug Sales Forecasting

STAT 443 - Time Series & Forecasting - Project Proposal

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Introduction

Pharmaceutical drug demand follows the underlying dynamics of diseases, which is influenced by the time-variant factors such as environmental, behavioural, and biological factors. The sales data provides an informative measure of disease dynamics at a larger scale. Prior research into the seasonality of respiratory infections, influenza, and allergic diseases, concludes that it is driven by environmental factors such as temperatures, humidity levels, and outdoor allergen levels like pollen or mold [1, 2]. This seasonality shows effect on the measurable fluctuation in medication demand [3]. Prescribed drugs for chronic conditions or lifestyle show different behaviour that is characterized by higher inertia and weaker seasonal effects. Moreover, similar ailments can be treated by different drug classes, where the choice may depend on the patient profile, severity of the condition, and more. Mild allergic reactions may lead to antihistamine use (R06), while more severe cases may require airway medications (R03). These clinical differences indicate that there might be meaningful cross-correlational patterns, and potentially useful variables for forecasting.

This project investigates whether Anatomical Therapeutic Chemical (ATC) classification system classified drugs exhibit distinct, interpretable patterns in sales volume and evaluates which time series models best capture these dynamics for forecasting.

Research Question

Do ATC classes of pharma drug sales exhibit disease-driven seasonal and cross-correlational patterns, and what factors explain these patterns? Can traditional time-series models (Holt-Winters, ARIMA) and multivariate regression with lagged features accurately capture the forecasting dynamics across different ATC drug classes?

Dataset Information

The [Pharma Drug Sales dataset](#) [4] is built from a dataset consisting of 600,000 transactional data collected in 6 years (period 2014-2019), indicating date and time of sale, pharmaceutical drug brand name and sold quantity, exported from the Point-of-Sale system from a retail drug store in *Niš, Serbia*.

The selected group of drugs from the dataset (57 drugs) is classified into eight of the categories in the ATC Classification System. The information in the dataset contains the daily sales volume of the pharmaceutical drugs from the categories mentioned in *Table 1*.

Variables

Variable Name	Data Type	Description
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datum	<i>Date-Time</i>	Represents the date of sales for the volume sold for each ATC drug class. This spans from January 1 st , 2014, to October 7 th , 2019.
M01AB	<i>Continuous</i>	Daily sales volume of M01AB drug class which includes the following substances: • Anti-inflammatory and antirheumatic products, non-steroids, Acetic acid derivatives and related substances
M01AE	<i>Continuous</i>	Daily sales volume of M01AE drug class which includes the following substances: • Anti-inflammatory and antirheumatic products, non-steroids, Propionic acid derivatives
N02BA	<i>Continuous</i>	Daily sales volume of N02BA drug class which includes the following substances: • Other analgesics and antipyretics, Salicylic acid and derivatives
N02BE	<i>Continuous</i>	Daily sales volume of N02BE drug class which includes the following substances: • Other analgesics and antipyretics, Pyrazolones and Anilides
N05B	<i>Continuous</i>	Daily sales volume of N05B drug class which includes the following substances: • Psycholeptics drugs, Anxiolytic drugs
N05C	<i>Continuous</i>	Daily sales volume of N05C drug class which includes the following substances: • Psycholeptics drugs, Hypnotics and sedatives drugs
R03	<i>Continuous</i>	Daily sales volume of R03 drug class which includes the following substances: • Drugs for obstructive airway diseases
R06	<i>Continuous</i>	Daily sales volume of R06 drug class which includes the following substances: • Antihistamines for systemic use

Table 1: Data variable Information

Time Series | Explanatory or Target Variables

Sales from each ATC drug class are treated as separate time series, with sales volume as the target variable. Forecasting models will primarily use lagged values of each series and may also incorporate lagged sales from related drug classes as explanatory variables to capture shared disease-driven dynamics. This allows comparison between univariate models and models that account for cross-series dependence.

References

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